Polymex Plant Pulse: why deliveries are late, and the fix

1 Jan 2026 to 31 Mar 2026 · for Hervé and the executive team

WHAT HERVÉ ASKED US TO SOLVE

On-time-in-full delivery has been slipping since January, when two retail clients started ordering small, frequent batches. He wants to see the plant on one page.

CEO HYPOTHESIS ✓ Supported by the shift logs

Small, frequent batches multiplied changeovers on extrusion, the plant's constraint. Every changeover minute there is lost plant output, and that lost output is what is making deliveries late.

Changeovers per day	Minutes per changeover	Capacity lost to changeovers	Good output
7.4 → 10.4	**102 → 99 min**	**29.0 → 39.7 t/day**	**233.6 → 223.9 t/day**
+40%, January to March	flat: the issue is how often, not how long	+10.7 t/day	−9.7 t/day

The shift logs don't include OTIF, so the link from lost output to late deliveries is inferred, not measured. First question for Hervé on Monday: what is OTIF by client since January, and is the order book above what we ship?

THE PRIZE

Prize, run-rate	Revenue capacity	Contribution	Plant OEE
+24.0 t/day	**€15.1M/yr**	**€3.0M/yr**	**58.2% → 64.5%**
8,774 t/yr of good output	10.2% of today's €148M	at an assumed 20% margin	March run-rate → target

LEVERS AND FINANCIAL VALUE

Lever (applied in order to the March run-rate)	Owner	Good t/day	t/yr	Revenue/yr	Contribution/yr
1. Batch policy for the two retail clients Bring changeovers back to the January rate of 7.4 a day (from 10.4) with minimum batch sizes or consolidation windows	Sales + planning	+9.3	3,381	€5.8M	€1.2M
2. SMED on extrusion: −30% minutes per changeover Cut the average changeover from 99 min to 70 min	Operations	+6.9	2,508	€4.3M	€0.9M
3. M02 breakdowns down to peer level M02 logs 231 breakdown min a day against a 109 min median on the other lines	Maintenance	+3.8	1,384	€2.4M	€0.5M
4. M07 speed up to peer level M07 runs at 77.6% of nameplate against a 88.3% median on the other lines	Process engineering	+4.1	1,501	€2.6M	€0.5M
Total		**+24.0**	**8,774**	**€15.1M**	**€3.0M**

Revenue at 1,721 €/t (€148M ÷ 86,000 t, from the brief). Contribution assumes a 20% margin, which still needs confirming. The gain fits the plant: mixing can feed +32.7 t/day more and packing can take +45.5 t/day more before either becomes the constraint. Extra tonnes only become revenue if the orders are there; late deliveries suggest they are.

MECE ISSUE TREE

Why is on-time-in-full delivery slipping since January?

✓ Supported **A. Extrusion, the constraint, delivers less good output**

Nameplate capacity 384.6 t/day splits into six parts that add up exactly. Only changeovers grew.

Where capacity goes (t/day)	Jan	Mar	Change	Q1 value/yr
A1 Availability: changeovers	**29.0**	**39.7**	**+10.7**	**€21.3M**
A1 Availability: breakdowns M02 is 2× its peers	35.0	34.5	−0.4	€21.5M
A1 Availability: other stops	34.3	34.2	−0.1	€21.4M
A2 Performance: speed loss M07 is slowest	39.3	37.7	−1.7	€24.2M
A3 Quality: scrap	15.3	14.6	−0.6	€9.4M
Good output	**233.6**	**223.9**	**−9.7**	**€143.6M**

✕ Rejected **B. Another stage limits the flow**

Mixing is blocked 11.7% of its time, packing is starved 16.5%, and extrusion loses only 0.6%.

? Untested **C. Enough output, but the wrong product at the wrong time**

Needs the production schedule against order due dates.

? Untested **D. Demand or promised lead times outgrew the plant**

Needs the order book and OTIF by client.

KPIS AND TARGETS

KPI	Now (Mar)	Target	Driven by
OTIF by client (%)	not in data	**set with Hervé**	Outcome KPI, needs order data
Good output (t/day)	223.9	**247.9**	All levers
Plant extrusion OEE (%)	58.2%	**64.5%**	All levers
Changeovers per day	10.4	**7.4**	1. Batch policy
Minutes per changeover (min)	99	**70**	2. SMED
Changeover share of planned time (%)	10.3%	**5.1%**	1 + 2
M02 availability (%, Q1)	65.9%	**74.4%**	3. Breakdowns
M07 performance (%, Q1)	77.6%	**88.3%**	4. Speed
Data anomalies per quarter	6	**0**	Fix logging at source

LEADERSHIP DECISIONS REQUIRED

1. **Sales director:** Agree a batch policy with the two retail clients: minimum batch sizes, consolidation windows or a small-batch price. This is the largest lever, worth €5.8M a year in revenue.
2. **Site director:** Fund SMED on the 7 extrusion lines and nowhere else. Mixing and packing are waiting on extrusion, so extra capacity there would add nothing.
3. **Planning:** Run a fixed product wheel on extrusion so the changeover count stays capped.
4. **Maintenance and process engineering:** Find the root cause of M02's breakdowns and M07's speed loss.
5. **Finance and sales:** Confirm the contribution margin, and share the order book and OTIF by client so the outcome can be measured.
6. **Data owner:** Fix the 6 logging anomalies at the source before these KPIs drive bonuses or capex.

Source: extrusion_shift_log.csv (1,890 rows) and stage_daily.csv, 6 anomalies corrected or flagged (see dashboard Data notes). Annualised good output 83,460 t vs 86,000 t in the brief. Assumed, not in the data: a 20% contribution margin (confirm with finance); SMED cuts minutes per changeover by 30% (low end of the typical 30 to 50%). Every number is computed from the CSVs at build time (2026-10-12).